

Yan Zhanglu

Research Fellow, National University of Singapore (NUS)

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Summary

Research Fellow at NUS working on **energy-efficient neuromorphic computing**, with a focus on **spiking neural networks (SNNs)**, **hardware-compatible algorithms**, and **hardware-level spike neuron design**. Recent publications span top venues such as Nature Communication, TPAMI, TCAD, TNNLS, ICLR, ICML, Acs Synth.Bio, WWW, ASPLOS, and AAAI.

Research Interests

Neuromorphic computing; spiking neural networks; quantization and mixed-precision inference; transformer/LLM-efficient inference; algorithm-hardware co-design.

Work Experience

National University of Singapore (NUS), Singapore

Oct 2024 – Present

Research Fellow, mentored by Prof. Weng-Fai Wong

- Neuromorphic algorithms and architecture-aware learning/inference for energy-efficient AI.
- LLM for biomedical and biological applications.

Education

National University of Singapore (NUS)

Ph.D., Computer Science, 2024

National University of Singapore (NUS)

M.Sc., Artificial Intelligence, 2020

Xi'an Jiaotong University (XJTU)

B.Sc., Computer Science, 2019

Selected Publications(First/Corresponding Author)

1. Kaiwen Tang, Di Yu, Jiaqi Zheng, Changze Lv, Qianhui Liu, Zhanglu Yan, Weng-Fai Wong. ShiftLIF: Efficient Multi-Level Spiking Neurons with Power-of-Two Quantization. (*ACM MM 2026, CCF-A*). *Corresponding author*.
2. Fanfan Li, Zhanglu Yan, Jiayi Mao, Guolei Liu, Huihui Ren, Bangbang Qin, Zhongfang Zhang, Haiyue Zhang, Yiyang Shen, Zeqi Zheng, Weilong Feng, Dingwei Li, Yingjie Tang, Saisai Wang, Yaochu Jin, Tao Luo, Weng-Fai Wong, Hong Wang, Bowen Zhu. Spinal-Inspired Artificial Tactile Interneuron with High-Order Burst Spiking for Intelligent Edge Interfaces. (*Nature Communications 2026, JCR-Q1, SCI-Q1 Top*). *Co-first author*.
3. Zhenyu Bai, Zhanglu Yan, Gao Bin, Tulika Mitra, Weng-Fai Wong. SparrowSNN: A hardware/software co-design for energy efficient ECG classification. (*IEEE TCAD 2026, CCF-A*). *Corresponding author*.
4. Zhanglu Yan, Jiayi Mao, Qianhui Liu, Fanfan Li, Tao Luo, Gang Pan, Bowen Zhu, Weng-Fai Wong. Otters: An Energy-Efficient Spiking Transformer via Optical Time-to-First-Spike Encoding. (*ICLR 2026, CCF-A*). *First/Corresponding author*.
5. Zhanglu Yan, Shida Wang, Kaiwen Tang, Zhenyu Bai, Weng-Fai Wong. HyperSNN: A new efficient and robust deep learning model for resource constrained control applications. (*ACC 2026, Oral, Tsinghua A, CAA-A*). *First author*.
6. Chenyu Wang, Zhanglu Yan, Zhi Zhou, Xu Chen, Weng-Fai Wong. Energy-Efficient and Dequantization-Free Q-LLMs: A Spiking Neural Network Approach to Salient Value Mitigation. (*WWW 2026, CCF-A*). *Co-corresponding author*.
7. Kaiwen Tang, Zhanglu Yan, Weng-Fai Wong. Sorbet: A Neuromorphic Hardware-Compatible Transformer-Based Spiking Language Model. (*ICML 2025, CCF-A*). *Corresponding author*.
8. Zhanglu Yan, Kaiwen Tang, Jun Zhou, Weng-Fai Wong. Low Latency Conversion of Artificial Neural Network Models to Rate-encoded Spiking Neural Networks. (*TNNLS 2025, JCR-Q1, SCI-Q1 TOP*). *First author*.

9. **Zhanglu Yan**, Weiran Chu, Yuhua Sheng, Kaiwen Tang, Shida Wang, Yanfeng Liu, Weng-Fai Wong. Integrating Deep Learning and Synthetic Biology: A Co-Design Approach for Enhancing Gene Expression via N-Terminal Coding Sequences. (*ACS Synth. Biol.* 2024, JCR-Q1, SCI-Q1). *Co-corresponding author, co-first author*.
10. **Zhanglu Yan**, Jun Zhou, Weng-Fai Wong. CQ+ Training: Minimizing Accuracy Loss in Conversion from CNNs to SNNs. (*IEEE TPAMI* 2023, CCF-A, JCR-Q1, SCI-Q1 Top). *Corresponding author, Co-first author*.
11. **Zhanglu Yan**, Shida Wang, Kaiwen Tang, Weng-Fai Wong. Efficient Hyperdimensional Computing. (*ECML* 2023, Oral, CORE A). *First author*.
12. **Zhanglu Yan**, Jun Zhou, Weng-Fai Wong. Near lossless transfer learning for spiking neural networks. (*AAAI* 2021, CCF-A). *Corresponding author, Co-first author*.
13. **Zhanglu Yan**, Jun Zhou, Weng-Fai Wong. Energy Efficient ECG Classification with Spiking Neural Network. (*Biomedical Signal Processing and Control* 2021, JCR-Q1, SCI-Q2 Top). *First author*. High citation (over 150).

Other Publications

1. **Zhanglu Yan**, Jiayi Mao, Kaiwen Tang, Fanfan Li, Gang Pan, Tao Luo, Bowen Zhu, Qianhui Liu, Weng-Fai Wong. Otters++: A Time-to-First-Spike Based Energy Efficient Optical Spiking Transformer. [arXiv:2606.13016](#), 2026. *First author*.
2. **Zhanglu Yan**, Kaiwen Tang, Zixuan Zhu, Zhenyu Bai, Qianhui Liu, Weng-Fai Wong. Matterhorn: Efficient Analog Sparse Spiking Transformer Architecture with Masked Time-To-First-Spike Encoding. [arXiv:2601.22876](#), 2026. *First author*.
3. Kaiwen Tang, Jiaqi Zheng, Yuze Jin, Yupeng Qiu, Guangda Sun, **Zhanglu Yan**, Weng-Fai Wong. SpikySpace: A Spiking State Space Model for Energy-Efficient Time Series Forecasting. [arXiv:2601.02411](#), 2026. *Corresponding author*.
4. Zhenyu Bai, Pranav Dangi, Rohan Juneja, Zhaoying Li, **Zhanglu Yan**, Huiying Lan, Tulika Mitra. A Data-Driven Dynamic Execution Orchestration Architecture. (*ASPLOS* 2026, CCF-A).
5. **Zhanglu Yan**, Zhenyu Bai, Weng-Fai Wong. Reconsidering the Energy Efficiency of Spiking Neural Networks. [arXiv:2409.08290](#), 2024. *First author*.
6. Kaiwen Tang, **Zhanglu Yan**, Weng-Fai Wong. OneSpike: Ultra-low latency spiking neural networks. (*IJCNN* 2024, CORE-B). *Corresponding author, co-first author*.
7. Hanyu Yang, **Zhanglu Yan**, Huacheng Zhu, Bowen Yan, Wei Chen, Daming Fan. Numerical Solution-Based Algorithm Assisted Development of a Continuous Flow Microwave Reactor. (*Chemical Engineering Journal* 2024, JCR-Q1).
8. **Zhanglu Yan**, Jun Zhou, Weng-Fai Wong. EEG Classification with Spiking Neural Network: Smaller, Better, More Energy Efficient. (*Smart Health* 2022). *First author*.

Academic Service

- **Area Chair:** ICIG
- **Conference Reviewer:** ICLR, ICML, AAAI, ICCAD.
- **Journal Reviewer:** IEEE TPAMI, TNNLS.

Last updated: July 08, 2026